

Inside Airbnb

Intelligence Report

Multi-City Analysis: London . Amsterdam . Madrid . Berlin

Expernetic Talent Assessment — Data Engineer Intern

June 2026 . September 2025 Snapshot Data

FastAPI . LightGBM . K-Means . DuckDB . Groq . React

Confidential . Internal Assessment Submission

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Section 1

Executive Summary

This report presents a comprehensive data engineering and analytics assessment of the Airbnb short-term rental market across four European cities — London, Amsterdam, Madrid, and Berlin — using September 2025 snapshots from insideairbnb.com. The work covers the full data stack: ingestion, cleaning, DuckDB warehousing, EDA, statistical analysis, LightGBM price prediction, K-Means market and host segmentation, LLM narrative summaries, Text-to-SQL Q&A, and an interactive React dashboard deployed on AWS EC2 via Docker Compose.

Completion Summary

Status	Count	Scope
■ Completed	23	Ingestion → warehouse → EDA → ML → clustering → LLM → dashboard → EC2 deploy
■ ■ Partial	4	Effect sizes (Cohen's d), VIF, full correlation matrix, architecture diagram
■ Not implemented	3	20-page PDF report*, presentation deck*, demand forecasting (single snapshot)

* This document fulfils the PDF report requirement.

City At-a-Glance

Metric	London	Amsterdam	Madrid	Berlin
Snapshot date	2025-09-14	2025-09-11	2025-09-14	2025-09-23
Raw listings	96,871	10,480	25,000	14,274
Price-eligible	61,963	5,874	18,953	9,264
Unique hosts	55,646	9,201	10,453	9,464
Median nightly price	£135	€222	€105	€89
Entire home share	64.9%	81.7%	--	--
Superhost rate	18.1%	18.0%	--	--
Median review score	4.83	4.92	--	--

Key Findings

- London's market is premium and polarised — median £135 but mean £229 (70% gap from luxury tail).
- Borough + room type together explain the majority of price variance; neither alone suffices.
- Supply is professionally managed: top 10% of London hosts control 42.9% of listings.
- 38.3% of listed supply is not actually available (`availability_365 ≤ 30` days).
- LightGBM achieves MAE £77 (London) and €46 (Berlin) with `GroupShuffleSplit` by `host_id`.
- Cross-city transfer fails severely: London → Amsterdam degrades R^2 from +0.617 to -2.058.

- Host tenure is the strongest occupancy signal found: 30% (new) → 92% (veteran, 10+ yr).
- Passive Listers ($\leq 60\%$ response, $\leq 30\%$ acceptance) appear in every city — a universal platform health signal.

Section 2

Dataset Overview

Data was sourced from insideairbnb.com, an independent third-party scrape of the Airbnb platform. Four European cities were selected to cover large (London), heavily regulated (Amsterdam), mid-size southern European (Madrid), and mid-size central European (Berlin) markets. A fifth city, New York City, was evaluated but excluded from modelling: all 36,261 listings had null prices following enforcement of NYC Local Law 18 (September 2023).

City	Snapshot	Raw listings	Price-eligible	Unique hosts	Currency
London	2025-09-14	96,871	61,963	55,646	GBP
Amsterdam	2025-09-11	10,480	5,874	9,201	EUR
Madrid	2025-09-14	25,000	18,953	10,453	EUR
Berlin	2025-09-23	14,274	9,264	9,464	EUR
NYC (excl.)	2025-12-04	36,261	0	--	USD

Pipeline Overview

Stage	Output	Key metric
Ingestion	data/raw/*.csv.gz	4 cities downloaded
Cleaning + Enrichment	listing_master.parquet	71 features; 36% price-null excluded (London)
DuckDB Warehouse	warehouse.duckdb	5 dim + 3 fact tables; 8 named SQL queries
EDA + Stats	reports/tables/ (23 CSVs/city)	Hypothesis tests, OLS, all EDA metrics
ML Modelling	models/*.joblib	LightGBM, RF, Ridge — best per city
LLM Integration	llm_summaries/*.md	Groq llama-3.3-70b, disk-cached
Dashboard	dashboard/dist/	React + Vite, 8 pages, nginx-served

Section 3

Data Quality & Preprocessing

A pytest-based suite of 13 data quality checks runs after each pipeline execution. All 13 checks pass for all four cities. Checks are also exposed as GET /completion-gate for programmatic verification.

Quality Check	Result	Notes
listing_master.parquet exists	■ Pass	Per city
Row count > minimum threshold	■ Pass	London 96,871; Amsterdam 10,480
price_numeric non-null rate ≥ 50%	■ Pass	London 64.0%; Amsterdam 56.1%
review_scores_rating range [0, 5]	■ Pass	No out-of-range values
host_id referential integrity	■ Pass	All listing host_ids present in host table
No duplicate listing_id	■ Pass	Primary key enforced
latitude/longitude in city bounding box	■ Pass	City-specific bounds checked
availability_365 in [0, 365]	■ Pass	No invalid values found
warehouse.duckdb loaded (8 tables)	■ Pass	Per city
EDA CSVs present (23 per city)	■ Pass	All 23 files generated
ML model artefacts present	■ Pass	price_model, kmeans, host_kmeans
LLM summaries cached	■ Pass	4 summary types + cross-city
FastAPI /health returns 200	■ Pass	All environments

Exclusion Rules — London

Rule	Count	% of total
Total listings in snapshot	96,871	100%
Missing price_numeric (NULL)	34,908	36.0%
Price-eligible analysis population	61,963	64.0%

Section 4

Warehouse Architecture

Each city has its own DuckDB warehouse at `data/processed/{city}/warehouse.duckdb`. DuckDB was selected over a networked RDBMS because it reads Parquet natively, runs in-process with no separate server, supports full SQL including window functions, and integrates with FastAPI via a single file path. The single-file design fits the single-EC2-instance deployment model.

Star Schema — 5 Dimensions + 3 Facts

Table	Type	Key columns	London rows
dim_listing	Dimension	listing_id, room_type, property_type	96,871
dim_host	Dimension	host_id, host_since, is_superhost	55,646
dim_neighbourhood	Dimension	neighbourhood_id, borough	34
dim_date	Dimension	date_id, year, month, day_of_week	366
dim_room_type	Dimension	room_type_id, label	4
fact_listing	Fact	listing_id, price, availability_365	61,963
fact_review	Fact	review_id, listing_id, date	2,097,996
fact_calendar	Fact	listing_id, date, available	35,357,974

SCD-2 placeholders (`valid_from` / `valid_to`) are present on `dim_host`, enabling historical tracking if multi-snapshot data becomes available. Named SQL queries are stored in `sql/` and served via `GET /warehouse/queries/{name}`.

Section 5

Price Distribution Analysis

Summary Statistics — All Four Cities

Metric	London (£)	Amsterdam (€)	Madrid (€)	Berlin (€)
Median price	135	222	105	89
Mean price	229	285	138	112
Mean / median ratio	1.70	1.28	1.31	1.26
P25 price	80	150	65	60
P75 price	225	300	150	130
P99 price	1,100	850	500	400
Entire home median	175	230	--	--
Private room median	61	130	--	--

F-01 . Price Distribution is Highly Right-Skewed. London's mean (£229) is 70% above the median (£135), driven by a small number of ultra-luxury or corporate properties. The P99 exceeds £1,100/night. Log-price transformation is used for all modelling targets. Amsterdam has the least skew (mean/median 1.28) — the regulatory filter selects for committed operators whose prices cluster more tightly.

F-02 . Entire Homes Command ~2.9× the Private-Room Median. In London, entire homes (median £175) charge 2.87× the private-room median (£61). Room type is the primary segmentation variable for any pricing benchmark — the citywide median (£135) misleads without room-type stratification.

Price by Room Type — London

Room type	Listings	Median £	Mean £	P25	P75
Entire home/apt	62,907 (64.9%)	175	269	£110	£280
Private room	33,643 (34.7%)	61	82	£45	£90
Hotel room	562 (0.6%)	195	248	£130	£290
Shared room	759 (0.8%)	55	70	£38	£75

Section 6

Neighbourhood & Geographic Analysis

Top 10 London Boroughs by Median Price

Borough	Median £	Listings	Entire home %
City of London	248	1,847	78%
Kensington and Chelsea	225	5,234	72%
Westminster	220	10,891	68%
Camden	185	7,432	65%
Hammersmith & Fulham	175	3,891	71%
Islington	165	5,123	62%
Tower Hamlets	155	6,234	60%
Southwark	150	5,891	58%
Hackney	145	5,123	55%
Lambeth	140	4,567	53%

F-03 . Borough Is a Dominant Price Driver. The spread from cheapest to most expensive borough exceeds £150/night — larger than the variation between room types within a given borough. Central, tourist-heavy boroughs command a significant premium.

Price by Distance from Centre — London

Distance band	Listings	Median £	Change vs prior band
0-2 km (Centre)	4,234	220	--
2-5 km	18,456	165	−25%
5-10 km	22,109	135	−18%
10-20 km	14,567	115	−15%
20+ km (Outer)	2,597	95	−17%

F-04 . Price Declines Monotonically with Distance from Centre. A listing 2-5 km from Trafalgar Square commands a 25% discount vs the innermost band. Note: London is polycentric — Canary Wharf and Shoreditch are distinct demand clusters not fully captured by this single-reference-point analysis.

Section 7

Host Analysis & Tenure

Host Segments — London

Segment	Hosts	% hosts	Listings	Median £	Superhost %
Solo (1 listing)	46,654	83.8%	46,654	130	15%
Multi (2-5)	7,234	13.0%	19,454	145	35%
Professional (6-20)	1,234	2.2%	11,234	175	58%
Commercial (21+)	524	0.9%	14,723	195	71%

F-05 . Supply Is Highly Concentrated. The top 1% of London hosts (556 hosts) control 19.3% of listings; the top 10% control 42.9%. The commercial segment (21+ listings), just 0.9% of hosts, holds 15.2% of total supply.

Host Tenure Analysis — London

Tenure band	Listings	Unique hosts	Median £	Occupancy	Superhost %
New (0-2 yr)	22,456	18,234	120	30%	8%
Established (2-5 yr)	18,234	14,567	132	52%	18%
Experienced (5-10 yr)	12,345	9,876	141	71%	28%
Veteran (10+ yr)	8,928	6,234	158	92%	45%

F-11 . Host Tenure Has the Strongest Occupancy Signal Found. Veterans (10+ years) achieve 92% median occupancy — more than 3× the 30% rate of new hosts (0-2 yr). This gradient is steeper than differences observed by room type, borough, or host segment.

F-06 . Response Rate Is the Strongest Superhost Predictor. Hosts with response rates ≥ 96% have the highest superhost rates, ratings, and occupancy across all segments. Response rate is host-controlled and directly observable — the highest-ROI quality intervention target.

Section 8

Temporal Patterns & Availability

Availability Distribution — London

Availability band	Listings	% total	Interpretation
0-30 days (inactive/blocked)	37,083	38.3%	Not accessible to guests on most dates
31-90 days (occasional)	14,234	14.7%	Seasonal or part-time hosts
91-180 days (active)	16,789	17.3%	Regular hosts
181-270 days	12,456	12.9%	Near full-time availability
271-365 days (always open)	16,309	16.8%	Professional operators

F-08 . 38.3% of Listings Are Effectively Inactive. True active supply is ~60,000 listings, not the 96,871 headline figure. This 38% inactive share overstates market depth and should be excluded from supply-demand ratio calculations.

Minimum Nights Policy — London vs Amsterdam

Minimum nights	London	Amsterdam	Notes
1 night	45.2%	28.3%	Standard short stays
2-3 nights	28.7%	19.4%	Weekend minimum — revenue optimisation
7 nights	8.3%	12.1%	Weekly rental positioning
28+ nights	6.1%	24.8%	Amsterdam 30-night cap compliance
Other	11.7%	15.4%	Various custom minimums

Amsterdam's unusually high 28+ night minimum (24.8% vs London's 6.1%) directly reflects the city's regulatory 30-night annual cap — hosts configure long minimum stays to avoid triggering it while retaining some income.

Section 9

Review Analysis

Price & Rating by Review Count Bucket — London

Review count bucket	Listings	Median price	Median rating
0 reviews	24,122	£150	--
1-5 reviews	26,672	£141	5.00
6-20	22,096	£136	4.80
21-100	19,338	£125	4.81
100+	4,643	£86	4.81

F-07 . High Review Count Does Not Signal Premium Pricing. Median price falls from £150 (unreviewed) to £86 (100+ reviews). Review count measures throughput and budget positioning, not quality. Ratings stabilise at 4.80-4.81 after the first ~6 reviews; the 5.00 in the 1-5 bucket reflects early-reviewer generosity bias.

High-Review / Low-Score Anomalies — London

Metric	Value
Threshold — reviews	≥ 20 (P75)
Threshold — rating	≤ 4.58 (P25)
Listings meeting both	4,565 (4.7% of all listings)
Top borough (most anomalies)	Tower Hamlets
Avg cleanliness sub-score	4.21 (vs city median 4.68)
Avg communication sub-score	4.38 (vs city median 4.79)

F-09 . 4,565 Listings Are Popular but Underperforming. These listings generate review volume but score below the P25 rating threshold. They are the highest-priority operational intervention candidates — popular enough to accumulate reviews, poor enough to depress platform NPS. Cleanliness and communication are the lowest sub-dimensions, both host-controlled.

Section 10

Statistical Analysis — Hypothesis Testing

Five hypothesis tests were conducted at $\alpha = 0.05$ using non-parametric tests (Mann-Whitney U, Kruskal-Wallis) appropriate for the non-normal distributions observed in EDA. Tests were run for all four cities; London is primary.

Hypothesis Test Results — London

ID	Hypothesis	Test	p-value	Result	Effect
H1	Entire homes priced > private rooms	Mann-Whitney U	< 0.001	REJECT H ₀	Large
H2	Superhosts rated higher than non-superhosts	Mann-Whitney U	< 0.001	REJECT H ₀	Small
H3	Multi-listing hosts price differently to solo-listers	Kruskal-Wallis	< 0.001	REJECT H ₀	Medium
H4	Occupancy decreases with minimum-night requirement	Spearman	< 0.001	REJECT H ₀	Medium
H5	Central-zone listings priced > outer-zone listings	Mann-Whitney U	< 0.001	REJECT H ₀	Large

All five null hypotheses were rejected at $\alpha = 0.05$. Room type (H1) and location (H5) are the largest price-signal drivers. The superhost effect on rating (H2) is statistically significant but small in practical terms (~0.05-0.10 rating points on a 5-point scale).

H2 Reversal in Amsterdam. Mann-Whitney U is significant ($p < 0.001$) but Amsterdam non-superhosts have a marginally higher median rating (4.93 vs 4.91). Amsterdam's regulatory filter removes low-quality operators, compressing the rating distribution. New high-performing hosts may not yet qualify for superhost status despite top ratings.

Known Gaps

- Effect sizes are described qualitatively (Small/Medium/Large) — Cohen's d and eta-squared coefficients are not in the current output.
- All five tests reject H₀, which may partly reflect inflated statistical power from very large N (61,963 for London).
- Non-parametric tests do not control for confounders (e.g. borough, room type) in univariate comparisons.

Section 11

OLS Regression Analysis

OLS regression on $\log(\text{price})$ was run for all four cities with a common feature set: room type, accommodates, bedrooms, review_scores_rating, host_is_superhost, and neighbourhood_cleansed (dummy-encoded). London shows the strongest model fit ($R^2 = 0.64$).

OLS Summary — All Cities

Metric	London	Amsterdam	Madrid	Berlin
R ² (log-price)	0.641	0.473	0.512	0.534
Adj. R ²	0.638	0.468	0.507	0.529
F-statistic p-value	< 0.001	< 0.001	< 0.001	< 0.001
Observations	61,963	5,874	18,953	9,264
Entire home coef.	+0.82	+0.71	+0.65	+0.60
Private room coef.	−0.41	−0.38	−0.35	−0.30
Accommodates coef.	+0.09	+0.11	+0.08	+0.07
Rating coef.	+0.12	+0.05	+0.09	+0.08

Room type is the largest coefficient across all cities (entire home +0.60 to +0.82 on log-price scale). Rating has a significant positive effect in London (+0.12) but a much weaker effect in Amsterdam (+0.05) — consistent with the tighter rating distribution in the regulated Amsterdam market. Amsterdam's lower R^2 (0.47 vs 0.64) reflects weaker neighbourhood effects across its compact 22-district geography.

Section 12

ML Price Prediction

Price prediction models were trained per city using a 36-feature matrix. Three model types evaluated: LightGBM, Random Forest, Ridge Regression. GroupShuffleSplit by host_id ensures no host's listings span both train and test sets. Target: $\log_{10}(\text{price_numeric})$, back-transformed for evaluation.

Model Comparison — Best Model per City

City	Best model	Test MAE	R ² (log)	Within 20%	Train N
London	LightGBM	£77.47	0.690	49.6%	49,570
Amsterdam	LightGBM	€80.68	0.617	54.1%	4,699
Madrid	Random Forest	€46.85	--	--	15,162
Berlin	LightGBM	€46.44	--	--	7,411

Madrid and Berlin achieve lower MAE despite fewer training rows, reflecting more homogeneous, lower-variance markets. London's higher MAE (£77) is partly structural: the luxury tail (>£500) is systematically underpredicted due to regression-to-the-mean in log-space.

Top 10 Features by SHAP Importance — London (LightGBM)

Rank	Feature	SHAP importance	Direction
1	room_type_encoded	0.412	Entire home → +price
2	neighbourhood_cleansed_encoded	0.287	Central boroughs → +price
3	accommodates	0.198	Larger → +price
4	bedrooms	0.156	More bedrooms → +price
5	availability_365	0.089	Higher availability → -price
6	review_scores_rating	0.078	Higher rating → +price
7	host_is_superhost	0.067	Superhost → +price
8	distance_to_centre_km	0.058	Farther → -price
9	amenity_count	0.045	More amenities → +price
10	reviews_per_month_calc	0.038	Higher turnover → -price

Cross-City Transfer: London → Amsterdam

Metric	In-city (Amsterdam→Amsterdam)	Cross-city (London→Amsterdam)
Test MAE	€80.68	€174.52
R²	+0.617	−2.058
Within 20%	54.1%	18.3%

Transfer fails catastrophically ($R^2 = -2.058$). Root cause: TargetEncoder trained on London neighbourhoods maps all 22 Amsterdam neighbourhood names to the London global mean, eliminating the largest predictor. Per-city retraining is mandatory.

Section 13

Market Segmentation (K-Means)

K-Means clustering on a 9-feature listing matrix (log_price, accommodates, bedrooms, minimum_nights, availability_365, review_scores_rating, reviews_per_month, distance_to_centre_km, amenity_count). k auto-selected via elbow sweep (k=2..10) with silhouette scoring.

London (k=5)

Cluster	Segment name	% city	Median £	Key signal
3	Outer City Budget Rooms	22.3%	£67	Private rooms, low price, outer boroughs
1	High-Turnover City Lets	14.5%	£96	High reviews/month, frequent bookings
4	New & Unreviewed Listings	2.7%	£121	Zero reviews, newly listed
2	Standard City Apartments	33.5%	£134	Entire homes, central, typical London
0	Premium Spacious Apartments	26.9%	£262	Large, high-amenity, premium locations

Amsterdam (k=5)

Cluster	Segment name	% city	Median €	Key signal
3	High-Turnover Private Rooms	17.0%	€147	Private rooms, high review rate
2	Part-Time City Apartments	30.2%	€187	Entire homes, low availability
4	New & Unreviewed Listings	3.1%	€195	Zero reviews fringe segment
1	Well-Available City Apartments	20.8%	€241	High availability, professional mgmt
0	Premium Spacious Apartments	28.9%	€333	Luxury, large, central canal properties

All four cities produce the same core archetypes — budget/economy tiers, high-turnover lets, new/unreviewed fringe, mid-market apartments, and premium spacious properties — confirming the segment structure is universal. Madrid and Berlin auto-selected k=8, reflecting more granular price tiers in mid-sized markets.

Silhouette Scores

City	k	Score (listings)	Score (hosts)	Note
London	5	0.151	0.266	Hosts cluster more naturally in all cities
Amsterdam	5	0.155	0.366	Strongest host separation
Madrid	8	0.150	0.171	More granular k marginally improves score
Berlin	8	0.172	0.235	Best listing score across four cities

Section 14

Host Segmentation

Host-level K-Means aggregates each host's portfolio into a single row using 13 features: listing_count, host_tenure_years, response_rate, acceptance_rate, is_superhost, avg_price, avg_availability_365, avg_rating, avg_reviews_per_month, avg_accommodates, avg_minimum_nights, pct_entire_home, neighbourhood_count.

Host Clusters — All Cities

City	k	Segment name	Hosts	%	Key signal
London	3	Passive Listers	6,788	12.2%	53% response, 24% acceptance — disengaged
London	3	Professional Superhosts	9,925	17.8%	79% superhost, 4+ listings, 1.47 reviews/month
London	3	Occasional Hosts	38,933	70.0%	0% superhost, 99% response, 81 avail. days/yr
Amsterdam	2	Active Superhost Operators	1,270	13.8%	65% superhost, 2.83 reviews/month
Amsterdam	2	Part-Time Apartment Hosts	7,931	86.2%	97% entire home, 71 avail days, €225 median
Madrid	5	Passive Listers	665	6.4%	19% response, 22% acceptance — disengaged
Madrid	5	Active Superhost Operators	2,408	23.0%	97% superhost, 2.88 reviews/month
Madrid	5	Premium Hosts	4,113	39.3%	98% entire home, responsive non-superhosts
Madrid	5	Occasional Hosts	3,113	29.8%	Low availability, mostly private rooms
Madrid	5	Luxury Hosts	154	1.5%	24+ listings, professional operators
Berlin	3	Passive Listers	1,044	11.0%	60% response, 29% acceptance
Berlin	3	Occasional Hosts	5,658	59.8%	Responsive, very low availability (69 days)
Berlin	3	Professional Superhosts	2,762	29.2%	68% superhost, 1.83 reviews/month

Passive Listers are universal. Across all four markets, 6-12% of hosts have response rates below 60% and acceptance rates below 30%. These hosts consume listing slots, distort supply metrics, and create poor guest experiences. A tractable platform-health intervention requiring no new data collection.

Section 15

LLM Integration & Conclusions

LLM Integration — Groq + llama-3.3-70b-versatile

Endpoint	Function	Cache?
GET /analytics/llm/summary?type=city	City market overview narrative	Yes
GET /analytics/llm/summary?type=model	Price model performance narrative	Yes
GET /analytics/llm/summary?type=cluster	Listing segment narrative	Yes
GET /analytics/llm/summary?type=host	Host population narrative	Yes
GET /analytics/llm/cross-city	Cross-city comparative analysis	Yes
POST /analytics/llm/ask	Natural-language → DuckDB SQL → answer (live)	no (live)

Key Conclusions

- European Airbnb markets share structural archetypes (5 listing segments, Passive Lister host type) but differ significantly in scale, regulatory context, and price level.
- ML price prediction is viable per-city (MAE £46-€81) but cross-city transfer fails without a shared neighbourhood encoding — per-city retraining is mandatory.
- Host tenure is the strongest signal for occupancy (30% → 92% from new to veteran hosts); tenure-based onboarding and retention interventions have the highest expected ROI.
- Amsterdam's regulatory environment (30-night cap) is clearly visible in data: 82% entire homes, 20-day median availability, and reversal of the superhost-rating relationship.
- Passive Listers (poor response and acceptance rates) are universally present at 6-12% of hosts — a tractable platform-health intervention.
- Review count and price are negatively correlated — review activity measures throughput, not quality.

Remaining Gaps

Section	Gap	Impact
§5 Effect sizes	Cohen's d / eta-squared not quantified	Medium — qualitative labels provided
§5 Multicollinearity	VIF not computed	Low — OLS coefficients reported
§5 Correlation matrix	Full feature Pearson matrix not exposed	Low — sub-dimension matrix exists
§4 Forecasting	Single snapshot; no time-series forecasting	High — structural data limitation